# Software Test Plan Document

**Project:** Vertex LMS v1.0 **Date:** November 2026 **Author:** Tlalane Sechele 202201293

## 1. Executive Summary & Purpose

This document defines the formal Software Test Plan for **Vertex LMS v1.0**, a C++17 standalone, offline-capable Learning Management System utilizing an embedded SQLite3 relational database engine. The purpose of this test plan is to specify the execution framework, quality goals, functional test matrix, and verification criteria necessary to guarantee architectural robustness, role-based access control enforcement, and database transaction consistency prior to system deployment.

## 2. Scope of Testing

- **Authentication & Security:** Validation of 16-byte random salt generation, SHA-256 equivalent password hashing, session management, and RBAC level enforcement (Student, Lecturer, HoD, Admin).
- **Relational Persistence & 3NF Integrity:** Foreign key constraint validation, cascading actions, transaction safety, and live database backup snapshotting.
- **Academic Workflow Business Logic:** Course allocation, student course roster enrollments, assignment binding, submission file fingerprinting, and class-wide deadline extensions (CR1).
- **Auditing & Governance:** Automated capturing of raw mark alterations with mandatory textual justification reasons written to an immutable audit trail, grade moderation workflows, and CSV export functionality.

## 3. Test Environment & Harness

All automated unit and integration tests are compiled using MSVC C++17 into a dedicated test execution harness ( `VertexLMSTests` ). The test harness provisions a isolated in-memory/temporary SQLite database file ( `test_vertex_lms.db` ) to ensure full independence from production records.

## 4. Comprehensive Test Case Matrix & Results

| Test ID | Subsystem | Test Description | Expected Outcome | Status |
|---|---|---|---|---|
| **TC-01** | AuthService | User Account Registration | User record stored with unique 16-byte salt and hashed password. | **PASS** |
| **TC-02** | AuthService | Credential Verification | Authenticates valid passwords and rejects invalid handles/passwords. | **PASS** |
| **TC-03** | CourseService | Course Creation & Allocation | Course code inserted uniquely and linked to target Lecturer ID. | **PASS** |

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|---|---|---|---|---|
| **TC-04** | CourseService | Student Enrollment | Enrollment recorded in junction table; duplicate enrollments blocked. | **PASS** |
| **TC-05** | AssignmentService | Assignment Creation | Assignment created with mandatory binding to valid Course ID. | **PASS** |
| **TC-06** | AssignmentService | Enrolled Solution Submission | Submission saved with generated file hash receipt. | **PASS** |
| **TC-07** | AssignmentService | Unenrolled Submission Block | Business logic rejects submissions from students not in course. | **PASS** |
| **TC-08** | SubmissionRepo | Grade Alteration Audit Trail | Marks updated to PROVISIONAL state and entry written to audit log. | **PASS** |
| **TC-09** | Database | Snapshot Backup Creation | Binary snapshot copy of SQLite database exported successfully. | **PASS** |